

ODORHEIUL SECUIESC, Romania

WMO: 151680

Lat: 46.2967N

Lon: 25.2917E

Elev: 524

StdP: 95.18

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.5	(c) -12.0	(d) -17.9	(e) 0.8	(f) -12.4	(g) -15.5	(h) 1.0	(i) -10.2	(j) 8.3	(k) 0.2	(l) 7.3	(m) -0.6	(n) 1.0	(o) 20	(p) 0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.5	(c) 30.3	(d) 19.5	(e) 28.8	(f) 18.9	(g) 27.4	(h) 18.5	(i) 20.7	(j) 27.9	(k) 20.0	(l) 26.9	(m) 19.2	(n) 25.8	(o) 2.0	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	14.2	23.1	17.7	13.5	22.6	17.0	12.9	21.6	62.4	28.0	59.6	26.7	57.1	26.1	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 6.0	(c) 4.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 7.5	(b) 6.0	(c) 4.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-2.5	0.5	4.0	9.3	14.6	18.5	19.3	19.7	14.7	10.1	5.3	-0.5		
	DBStd	8.69	4.86	4.60	4.49	4.73	3.30	3.02	2.86	2.96	4.37	3.63	4.48	4.59		
	HDD10.0	1451	387	266	191	70	6	0	0	0	8	47	151	325		
	HDD18.3	3411	646	499	444	271	121	34	24	21	122	255	392	583		
	CDD10.0	1257	0	1	6	49	148	254	287	302	151	50	10	0		
	CDD18.3	176	0	0	0	0	5	39	53	64	14	0	0	0		
	CDH23.3	1832	0	0	1	19	91	396	489	645	183	9	0	0		
	CDH26.7	478	0	0	0	0	8	89	133	206	43	0	0	0		
(14) Wind	WSAvg	1.6	1.2	1.5	1.8	2.1	1.8	1.6	1.7	1.6	1.7	1.5	1.5	1.3		
(15) Precipitation	PrecAvg	650	28	30	36	56	77	97	87	66	57	51	36	43		
	PrecMax	790	53	60	74	112	142	184	158	114	124	101	71	117		
	PrecMin	488	13	8	7	17	18	30	38	19	3	2	0	7		
	PrecStd	89	12	13	19	24	30	37	33	28	31	27	17	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	14.9	21.2	25.6	27.8	30.6	31.8	33.0	29.9	24.6	19.4	12.1		
		MCWB	6.5	9.3	11.2	14.1	15.8	20.9	20.0	20.1	18.1	14.7	11.9	8.0		
	2%	DB	7.2	11.8	17.3	23.3	26.0	29.0	29.9	30.8	27.6	22.5	16.9	8.8		
		MCWB	4.5	6.7	9.2	13.1	15.7	19.5	19.7	19.1	16.9	13.9	11.1	6.3		
	5%	DB	5.0	9.4	14.2	20.9	24.0	27.4	28.1	28.9	25.2	20.0	14.4	6.5		
		MCWB	2.9	5.5	8.0	12.2	15.5	18.8	19.1	18.9	16.1	13.0	9.6	4.6		
	10%	DB	3.2	6.8	11.0	18.1	21.8	25.6	26.2	27.1	22.8	17.1	12.0	4.7		
		MCWB	1.7	4.1	6.2	11.2	14.1	18.0	18.5	18.8	15.5	11.8	8.5	3.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	9.5	11.7	14.9	18.3	22.1	21.4	21.5	19.7	15.5	13.0	8.6		
		MCDB	9.4	14.1	20.4	23.4	24.1	29.7	28.4	29.3	28.2	22.5	18.1	11.0		
	2%	WB	4.6	7.4	9.9	13.9	16.9	20.5	20.5	20.7	18.0	14.6	11.7	6.5		
		MCDB	6.7	11.2	16.4	21.8	23.3	27.5	27.2	28.3	24.6	21.0	15.8	8.7		
	5%	WB	3.1	5.9	8.3	12.8	15.9	19.4	19.8	19.8	16.9	13.5	10.4	4.7		
		MCDB	4.7	8.7	13.5	19.9	22.2	26.0	26.2	26.9	23.5	18.9	14.0	6.2		
	10%	WB	1.9	4.3	6.9	11.6	15.0	18.6	19.1	19.0	16.0	12.2	8.9	3.5		
		MCDB	3.0	6.5	10.9	17.4	20.4	24.5	25.1	26.3	22.1	16.4	12.0	4.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	8.8	10.2	11.3	11.3	12.0	11.7	12.5	11.6	11.3	8.7	6.0		
		MCDBR	8.5	11.8	15.9	16.4	14.7	14.6	14.5	15.4	15.6	17.1	12.7	8.8		
	5% WB	MCWBR	5.9	7.3	8.1	7.6	6.7	6.4	6.0	6.0	6.3	9.0	7.3	6.3		
		MCWBR	7.6	10.9	13.8	14.8	13.0	13.7	12.9	13.9	13.8	15.4	11.2	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.331	0.373	0.428	0.408	0.405	0.425	0.432	0.391	0.354	0.330	0.304		
	taud		2.327	2.277	2.236	2.121	2.247	2.287	2.238	2.215	2.299	2.393	2.402	2.366		
	Ebn at Noon		785	832	845	826	860	862	838	814	818	799	748	741		
	Edh at Noon		76	98	119	146	133	129	134	132	111	87	71	66		
(44) All-Sky Solar Radiation	RadAvg		1.32	2.16	3.27	4.35	5.36	5.80	5.78	5.30	3.79	2.60	1.48	1.01		
	RadStd		0.16	0.24	0.37	0.44	0.49	0.50	0.49	0.47	0.41	0.32	0.18	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)		+0.67	N/A	N/A	+0.83	N/A	N/A	-146	-201	+81	+19

Nomenclature: See separate page